**EXERCISE 1** Use radians to find the arc length and area of a sector of a circle of :

1. radius 9 cm and angle $\frac{7\pi}{4}$
2. radius 4.93 cm and angle 4.67 radians

**EXERCISE 2** A sector has an angle of $107.9°$ and an arc length of 5.92 m. Find its :

1. radius
2. area

**EXERCISE 3** A sector has an angle of 1.19 radians and an area of 20.8 cm$^2$. Find its :

1. radius
2. perimeter

**EXERCISE 4** Find, in radians, the angle of a sector of :

1. radius 4.3 m and arc length 2.95 m
2. radius 10 cm and area 30 cm$^2$

**EXERCISE 5** Find $\theta$ (in radians) for each of the following, and hence find the area of each figure :

1. radius 6 cm, arc length 8 cm
2. radius 5 cm, arc length 8.4 cm
3. radius 8 cm, arc length 31.7 cm

**EXERCISE 6** For each unit circle illustrated :

1. state the exact coordinates of points $A$, $B$, and $C$ in terms of sine and cosine ;
2. use your calculator to give the coordinates of $A$, $B$, and $C$ correct to 3 significant figures.

**EXERCISE 7** With the aid of a unit circle, complete the following table :

| $\theta$ (degrees) | $0°$ | $90°$ | $180°$ | $270°$ | $360°$ | $450°$ |
|---|---|---|---|---|---|---|
| $\theta$ (radians) | | | | | | |
| $\sin\theta$ | | | | | | |
| $\cos\theta$ | | | | | | |
| $\tan\theta$ | | | | | | |

**EXERCISE 8**

**1.** Use your calculator to evaluate :

(i) $\sin 100°$ (ii) $\sin 80°$ (iii) $\sin 120°$ (iv) $\sin 60°$

(v) $\sin 150°$ (vi) $\sin 30°$ (vii) $\sin 45°$ (viii) $\sin 135°$.

**2.** Use the results from (a) to copy and complete : $\sin(180° - \theta) = \ldots$

**3.** Justify your answer using a diagram.

**4.** Find the obtuse angle with the same sine as :

(i) $45°$ (ii) $51°$ (iii) $\frac{\pi}{3}$ (iv) $\frac{\pi}{6}$.

**EXERCISE 9**

**1.** Use your calculator to evaluate :

(i) $\cos 70°$ (ii) $\cos 110°$ (iii) $\cos 60°$ (iv) $\cos 120°$

(v) $\cos 25°$ (vi) $\cos 155°$ (vii) $\cos 80°$ (viii) $\cos 100°$.

**2.** Use the results from (a) to copy and complete : $\cos(180° - \theta) = \ldots$

**3.** Justify your answer using a diagram.

**4.** Find the obtuse angle which has the *negative* cosine of :

(i) $40°$ (ii) $19°$ (iii) $\frac{\pi}{5}$ (iv) $\frac{2\pi}{5}$.

**EXERCISE 10** In which quadrants are the following true ?

(i) $\cos\theta$ is positive. (ii) $\cos\theta$ is negative.

(iii) $\cos\theta$ and $\sin\theta$ are both negative. (iv) $\cos\theta$ is negative and $\sin\theta$ is positive.

**EXERCISE 11** Complete :

| $\theta$ | $\sin\theta$ | $\sin(-\theta)$ | $\cos\theta$ | $\cos(-\theta)$ |
|---|---|---|---|---|
| 0.75 | | | | |
| 1.772 | | | | |
| 3.414 | | | | |
| 6.25 | | | | |
| $-1.17$ | | | | |

**EXERCISE 12** Find the possible exact values of $\cos\theta$ for :

**1.** $\sin\theta = \frac{1}{2}$

**2.** $\sin\theta = -\frac{1}{3}$

**3.** $\sin\theta = 0$

**4.** $\sin\theta = -1$

**EXERCISE 13** Find the possible exact values of $\sin\theta$ for :

**1.** $\cos\theta = \frac{4}{5}$

**2.** $\cos\theta = -\frac{3}{4}$

**3.** $\cos\theta = 1$

**4.** $\cos\theta = 0$

**EXERCISE 14** Without using a calculator, find :

**1.** $\sin\theta$ if $\cos\theta = \frac{2}{3}$ and $0 < \theta < \frac{\pi}{2}$

**2.** $\cos\theta$ if $\sin\theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$

**3.** $\cos\theta$ if $\sin\theta = -\frac{3}{5}$ and $\frac{3\pi}{2} < \theta < 2\pi$

**4.** $\sin\theta$ if $\cos\theta = -\frac{5}{13}$ and $\pi < \theta < \frac{3\pi}{2}$

**EXERCISE 15**

**1.** If $\sin x = \frac{1}{3}$ and $\frac{\pi}{2} < x < \pi$, find $\tan x$ exactly.

**2.** If $\cos x = \frac{1}{5}$ and $\frac{3\pi}{2} < x < 2\pi$, find $\tan x$ exactly.

**3.** If $\sin x = -\frac{1}{\sqrt{3}}$ and $\pi < x < \frac{3\pi}{2}$, find $\tan x$ exactly.

**4.** If $\cos x = -\frac{3}{4}$ and $\frac{\pi}{2} < x < \pi$, find $\tan x$ exactly.

**EXERCISE 16** Find exact values for $\sin x$ and $\cos x$ given that :

**1.** $\tan x = \frac{2}{3}$ and $0 < x < \frac{\pi}{2}$

**2.** $\tan x = -\frac{4}{3}$ and $\frac{\pi}{2} < x < \pi$

**3.** $\tan x = \frac{\sqrt{5}}{3}$ and $\pi < x < \frac{3\pi}{2}$

**4.** $\tan x = -\frac{12}{5}$ and $\frac{3\pi}{2} < x < 2\pi$

**EXERCISE 17** Without using a calculator, evaluate :

a) $\sin^2 60°$ b) $\sin 30° \cos 60°$ c) $4 \sin 60° \cos 30°$

d) $1 - \cos^2\left(\frac{\pi}{6}\right)$ e) $\sin^2\left(\frac{2\pi}{3}\right) - 1$ f) $\cos^2\left(\frac{\pi}{4}\right) - \sin\left(\frac{7\pi}{6}\right)$

g) $\sin\left(\frac{3\pi}{4}\right) - \cos\left(\frac{5\pi}{4}\right)$ h) $1 - 2\sin^2\left(\frac{7\pi}{6}\right)$ i) $\cos^2\left(\frac{5\pi}{6}\right) - \sin^2\left(\frac{5\pi}{6}\right)$

j) $\tan^2\left(\frac{\pi}{3}\right) - 2\sin^2\left(\frac{\pi}{4}\right)$ k) $2\tan\left(-\frac{5\pi}{4}\right) - \sin\left(\frac{3\pi}{2}\right)$ l) $\frac{2\tan 150^\circ}{1 - \tan^2 150^\circ}$

**EXERCISE 18** Find all angles between $0$ and $4\pi$ with :

a) a cosine of $\frac{\sqrt{3}}{2}$, b) a sine of $-\frac{1}{2}$, c) a sine of $-1$.

**EXERCISE 19**

Find $\theta$ if $0 \leq \theta \leq 2\pi$ and :

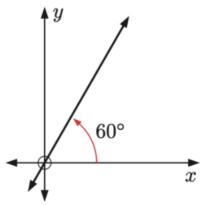
a) $\cos\theta = \frac{1}{2}$, b) $\sin\theta = \frac{\sqrt{3}}{2}$, c) $\cos\theta = -1$,

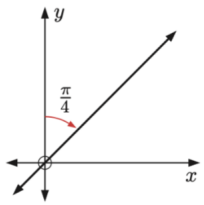
d) $\sin\theta = 1$, e) $\cos\theta = -\frac{1}{\sqrt{2}}$, f) $\sin^2\theta = 1$,

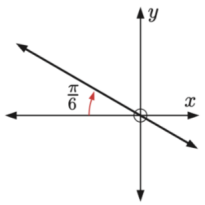
g) $\cos^2\theta = 1$, h) $\cos^2\theta = \frac{1}{2}$, i) $\tan\theta = -\frac{1}{\sqrt{3}}$, j) $\tan^2\theta = 3$.

**EXERCISE 20**

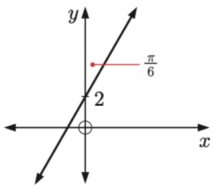
1 Find the equation of each line:

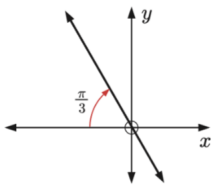
a 

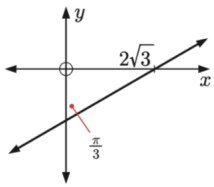

b 


c 


2 Find the equation of each line:

a 


b 


c 

**EXERCISE 21** Find $\frac{dy}{dx}$ **for :**

1. $y = \sin(2x)$
2. $y = \sin x + \cos x$
3. $y = \cos(3x) - \sin x$
4. $y = \sin(x+1)$
5. $y = \cos(3-2x)$
6. $y = \tan(5x)$
7. $y = \sin\left(\frac{x}{2}\right) - 3\cos x$
8. $y = 3\tan(\pi x)$
9. $y = 4\sin x - \cos(2x)$

**EXERCISE 22** Differentiate with respect to $x$ :

1. $x^2 + \cos x$
2. $\tan x - 3\sin x$
3. $e^x \cos x$
4. $e^{-x}\sin x$
5. $\ln(\sin x)$
6. $e^{2x}\tan x$
7. $\sin(3x)$
8. $\cos\left(\frac{x}{2}\right)$
9. $3\tan(2x)$
10. $x\cos x$
11. $\frac{\sin x}{x}$
12. $x\tan x$

**EXERCISE 23** Differentiate with respect to $x$ :

1. $\sin(x^2)$
2. $\cos(\sqrt{x})$
3. $\sqrt{\cos x}$
4. $\sin^2 x$
5. $\cos^3 x$
6. $\cos x \sin(2x)$
7. $\cos(\cos x)$
8. $\cos^3(4x)$
9. $\frac{1}{\sin x}$
10. $\frac{1}{\cos(2x)}$
11. $\frac{2}{\sin^2(2x)}$
12. $\frac{8}{\tan^3\left(\frac{x}{2}\right)}$

**EXERCISE 24** Find the gradient of the tangent to :

1. $f(x) = \sin^3 x$ at the point where $x = \frac{2\pi}{3}$.
2. $f(x) = \cos x \sin x$ at the point where $x = \frac{\pi}{4}$.